Local Moran's I: three extreme cases

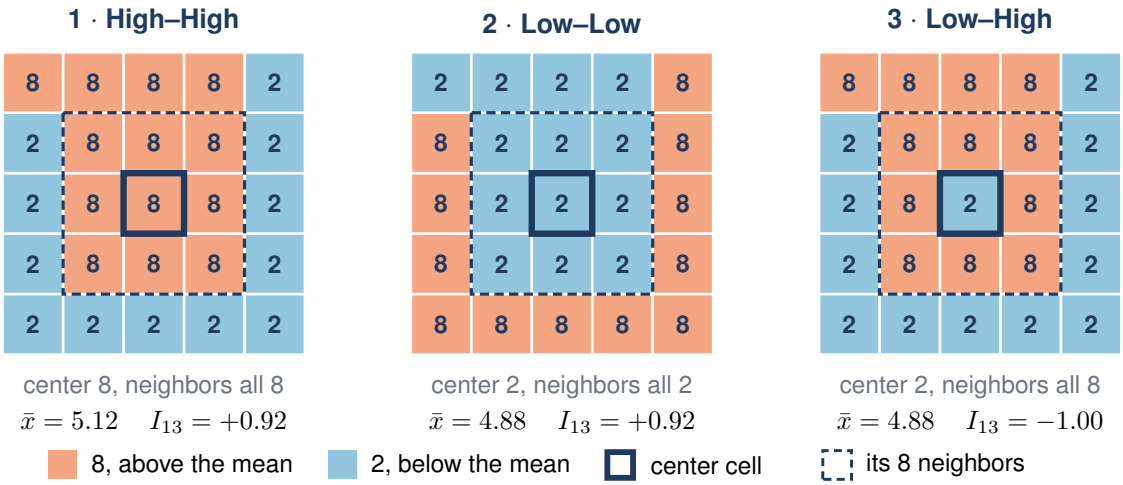
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Each grid below is a study area of 25 cells, with values of 2 or 8. We look at the center cell and its eight queen neighbors. Counting row by row from the top left, the center is cell 13.

With row-standardized weights, the local Moran's I of cell i is

$$I_i = \frac{z_i}{m_2} \sum_j w_{ij} z_j, \qquad z_i = x_i - \bar{x}, \qquad m_2 = \frac{1}{n} \sum_i z_i^2.$$

The sum $\sum_j w_{ij} z_j$ is simply the average deviation of the eight neighbors. In words: **(my deviation) $\times$ (my neighbors' average deviation)**, scaled by the variance.



Pattern	Center	Eight neighbors	Study-area mean $\bar{x}$	Local Moran's I_{13}
High–High	8	all = 8	5.12	$+0.92$
Low–Low	2	all = 2	4.88	$+0.92$
Low–High	2	all = 8	4.88	-1.00

Class question

The High–High and Low–Low grids give exactly the same I_{13}, although one center holds a high value and the other a low value. Why? What does a positive local Moran's I tell us, and what does it not tell us?

Two cautions. These are illustrative patterns. Before calling a cell a cluster or an outlier, we need a significance test, usually by permutation (`localmoran_perm()`). Local Moran's I is also **not bounded** by -1 and $+1$: the value of exactly -1 in the Low–High grid is a coincidence of the chosen values.

R code

```
library(spdep)

# 5 x 5 grid, queen neighbors, row-standardized weights
nb <- cell2nb(5, 5, type = "queen")
lw <- nb2listw(nb, style = "W")

# one line per grid row; the center is cell 13
values_HH <- c(8, 8, 8, 8, 2,
               2, 8, 8, 8, 2,
               2, 8, 8, 8, 2,
               2, 8, 8, 8, 2,
               2, 2, 2, 2, 2)

values_LL <- c(2, 2, 2, 2, 8,
               8, 2, 2, 2, 8,
               8, 2, 2, 2, 8,
               8, 2, 2, 2, 8,
               8, 8, 8, 8, 8)

values_LH <- c(8, 8, 8, 8, 2,
               2, 8, 8, 8, 2,
               2, 8, 2, 8, 2,          # low center
               2, 8, 8, 8, 2,
               2, 2, 2, 2, 2)

result_HH <- localmoran(values_HH, lw)
result_LL <- localmoran(values_LL, lw)
result_LH <- localmoran(values_LH, lw)

# local Moran's I at the center
result_HH[13, "Ii"]                     # 0.923
result_LL[13, "Ii"]                     # 0.923
result_LH[13, "Ii"]                     # -1

# quadrant: where the label comes from
attr(result_HH, "quadr")[13, "mean"]    # High-High
attr(result_LL, "quadr")[13, "mean"]    # Low-Low
attr(result_LH, "quadr")[13, "mean"]    # Low-High

# the same I by hand (High-High)
z     <- values_HH - mean(values_HH)    # deviations
m2    <- mean(z * z)                    # variance, 8.99
lag_z <- lag.listw(lw, z)               # neighbors' average deviation
z[13] * lag_z[13] / m2                  # 0.923
```

Answer to the class question

I_i is the product of two deviations. In High–High both are positive; in Low–Low both are negative. Either way, the product is positive.

	z_{13}	neighbors' average z	m_2	I_{13}
High–High	$+2.88$	$+2.88$	8.99	$(+2.88)(+2.88)/8.99 = +0.92$
Low–Low	-2.88	-2.88	8.99	$(-2.88)(-2.88)/8.99 = +0.92$
Low–High	-2.88	$+3.12$	8.99	$(-2.88)(+3.12)/8.99 = -1.00$

A positive I_i means **similarity** between a cell and its neighbors, not high values. The two values are identical, not only of the same sign, because the Low–Low grid is the High–High grid with 2 and 8 swapped: every deviation changes sign, and m_2 does not change. To tell High–High from Low–Low, we need the sign of z_i itself: the quadrant of the Moran scatterplot.